L2: Markov Decision Processes

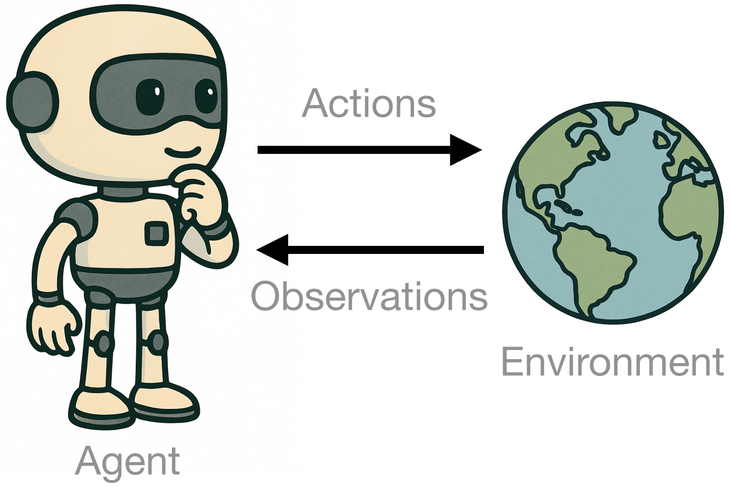
EECE 576N | From MDPs to Deep RL | Fall 2026

Cyrus Neary | cyrus.neary@ubc.ca

Learning Objectives

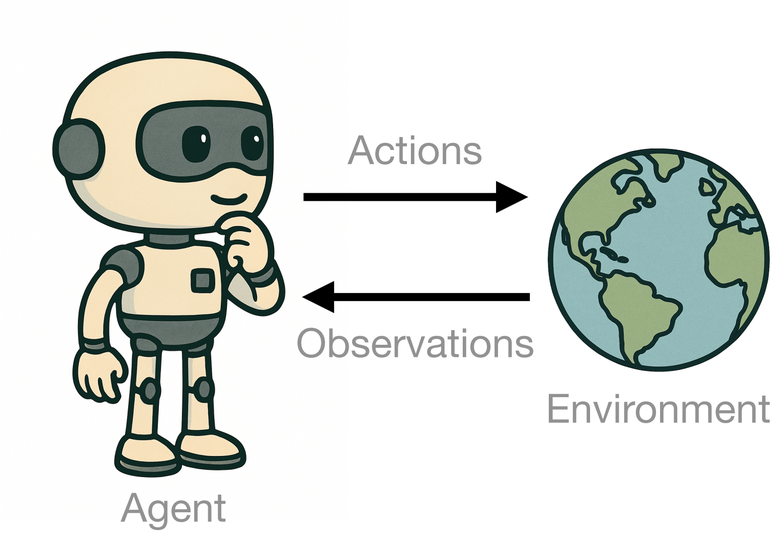
By the end of today's lecture, you should be able to…

- Explain the core modeling components of Markov decision processes (MDPs).
- Differentiate between classes of decision-making policies.
- Understand the components of the typical optimization objective in MDP problems.
- Define value functions and optimal value functions, and explain their role in sequential decision-making problems.



Sequential Decision-Making Problems

Objective: The decision-making *agent* should implement a *policy* that selects actions a_t at every *decision epoch* t to maximize some function of the resulting rewards $r_0, r_1, \ldots$

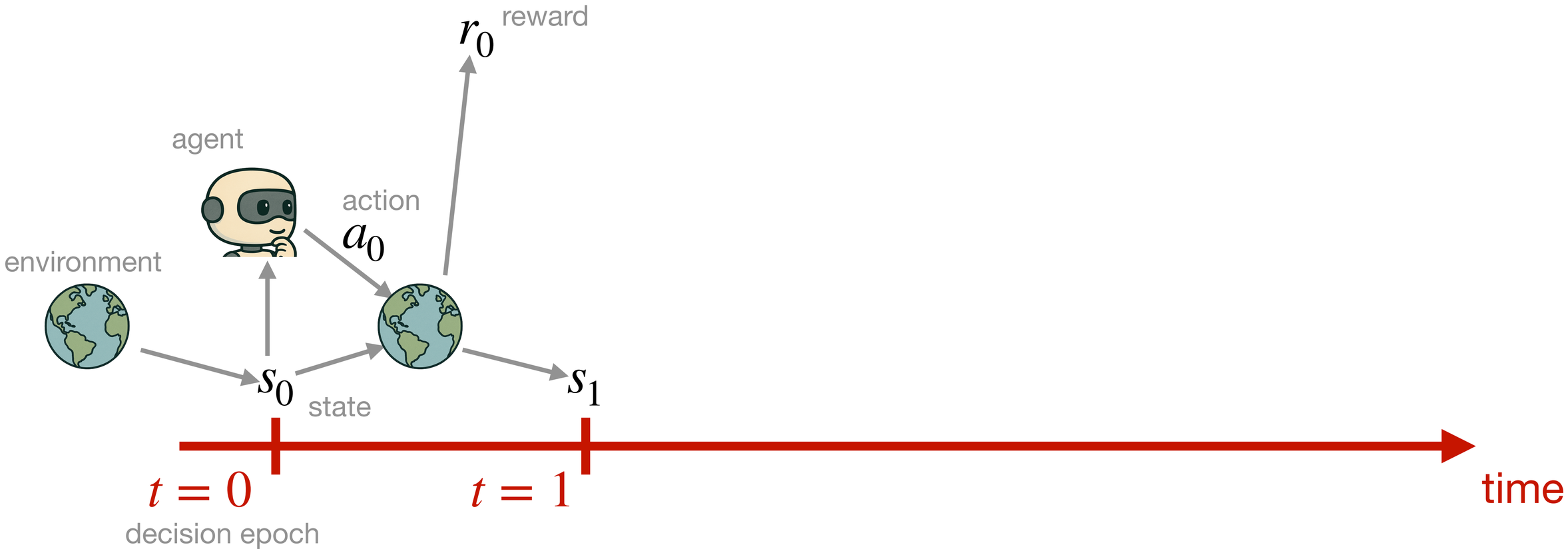


Sequential Decision-Making Problems

Objective: The decision-making *agent* should implement a *policy* that selects actions a_t at every *decision epoch* t to maximize some function of the resulting rewards $r_0, r_1, \ldots$

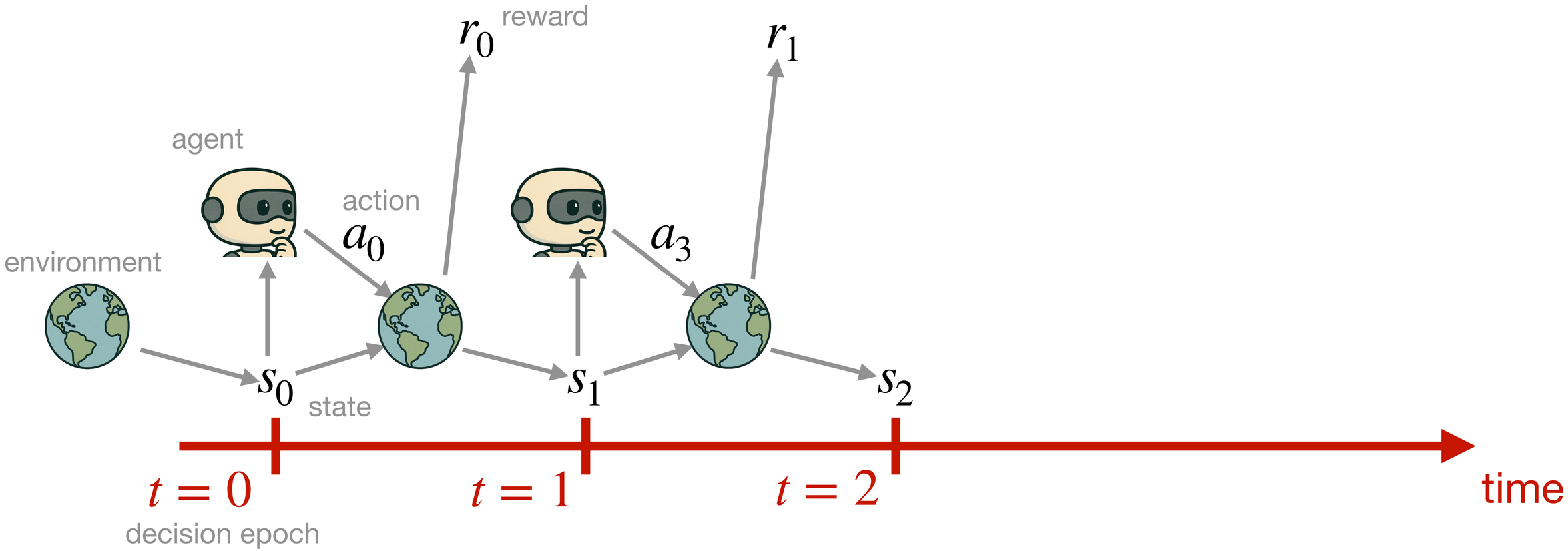
Sequential Decision-Making Problems

Objective: The decision-making *agent* should implement a *policy* that selects actions a_t at every *decision epoch* t to maximize some function of the resulting rewards $r_0, r_1, \ldots$



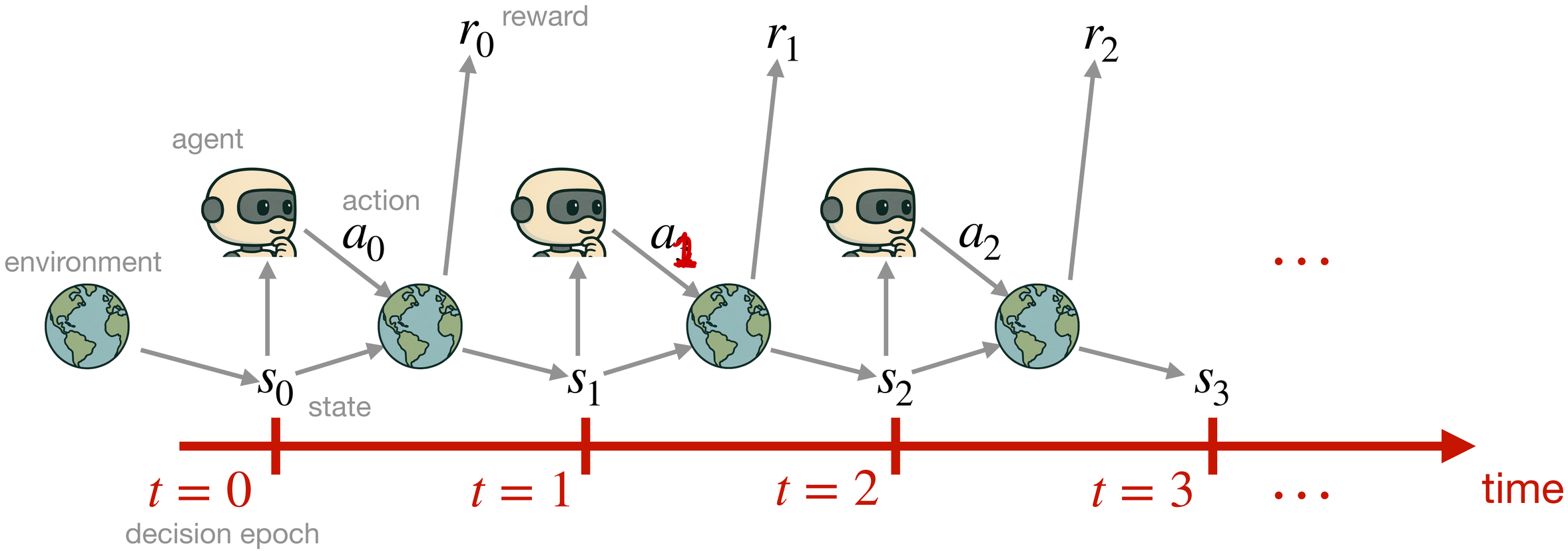
Sequential Decision-Making Problems

Objective: The decision-making *agent* should implement a *policy* that selects actions a_t at every *decision epoch* t to maximize some function of the resulting rewards $r_0, r_1, \ldots$

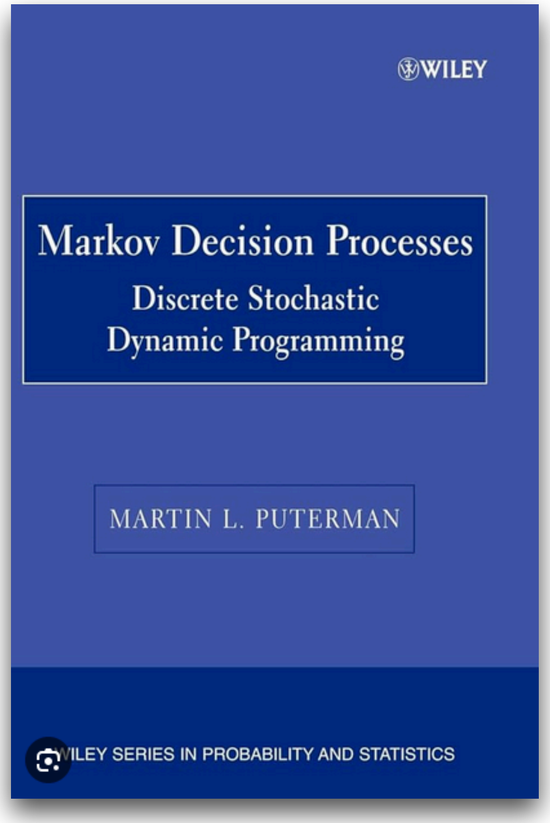


Sequential Decision-Making Problems

Objective: The decision-making *agent* should implement a *policy* that selects actions a_t at every *decision epoch* t to maximize some function of the resulting rewards $r_0, r_1, \ldots$



Modeling Sequential Decision-Making Problems



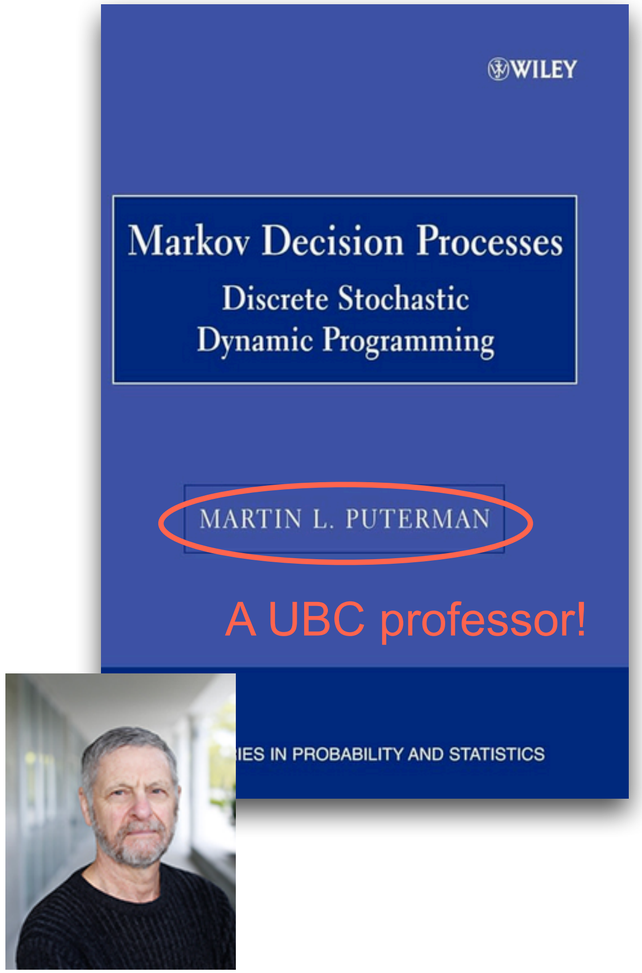
“Each day people make many decisions; decisions which have both immediate and long-term consequences. Decisions must not be made in isolation; today's decision impacts on tomorrow's and tomorrow's on the next day’s…

This book presents and studies a model for sequential decision making under uncertainty, which takes into account both the outcomes of current decisions and future decision making opportunities. While this model may appear quite simple, it encompasses a wide range of applications and has generated a rich mathematical theory.”

Puterman ML. Markov decision processes: discrete stochastic dynamic programming. John Wiley & Sons; 2014 Aug 28.

Today’s lecture draws from chapters 2, 3, and 5

Modeling Sequential Decision-Making Problems



"Each day people make many decisions; decisions which have both immediate and long-term consequences. Decisions must not be made in isolation; today's decision impacts on tomorrow's and tomorrow's on the next day's…

This book presents and studies a model for sequential decision making under uncertainty, which takes into account both the outcomes of current decisions and future decision making opportunities. While this model may appear quite simple, it encompasses a wide range of applications and has generated a rich mathematical theory."

Puterman ML. Markov decision processes: discrete stochastic dynamic programming. John Wiley & Sons; 2014 Aug 28.

Today's lecture draws from chapters 2, 3, and 5

Markov Decision Processes

$$\mathscr{M} = (S, A, T, R, \gamma, \mu)$$

Markov Decision Processes

$$\mathscr{M} = (S, A, T, R, \gamma, \mu)$$

S - States

Set of all possible states from which the agent will make decisions.

Markov Decision Processes

A - Actions

Set of all possible actions the agent could take.

S - States

Set of all possible states from which the agent will make decisions.

$$\mathscr{M} = (S, A, T, R, \gamma, \mu)$$

Markov Decision Processes

R- Reward function

$R(s, a) \in \mathbb{R}$ — A function defining the reward the agent receives for transitioning from state $s \in S$ to $s' \in S$ under action $a \in A$.

A - Actions

Set of all possible actions the agent could take.

$$\mathscr{M} = (S, A, T, R, \gamma, \mu)$$

$R(s,a,s')$

$R(s,a) = \mathbb{E}\left[\sum_{s'} T(s'|s,a) R(s,a,s')\right]$

S - States

Set of all possible states from which the agent will make decisions.

Markov Decision Processes

R- Reward function

$R(s, a) \in \mathbb{R}$ — A function defining the reward the agent receives for transitioning from state $s \in S$ to $s' \in S$ under action $a \in A$.

A - Actions

Set of all possible actions the agent could take.

$$\mathscr{M} = (S, A, T, R, \gamma, \mu)$$

μ - Initial state distribution

$\mu(s)$ denotes the probability that the initial state is s.

S - States

Set of all possible states from which the agent will make decisions.

Markov Decision Processes

R- Reward function

$R(s, a) \in \mathbb{R}$ — A function defining the reward the agent receives for transitioning from state $s \in S$ to $s' \in S$ under action $a \in A$.

A - Actions

Set of all possible actions the agent could take.

$$\mathscr{M} = (S, A, T, R, \gamma, \mu)$$

μ - Initial state distribution

$\mu(s)$ denotes the probability that the initial state is s.

S - States

Set of all possible states from which the agent will make decisions.

γ - Discount factor

$\gamma \in [0,1]$ — a modeling parameter that reduces the impact of future rewards on decisions made now.

Markov Decision Processes

$\mathscr{M} = (S, A, T, R, \gamma, \mu)$

R- Reward function

$R(s, a) \in \mathbb{R}$ — A function defining the reward the agent receives for transitioning from state $s \in S$ to $s' \in S$ under action $a \in A$.

A - Actions

Set of all possible actions the agent could take.

μ - Initial state distribution

$\mu(s)$ denotes the probability that the initial state is s.

S - States

Set of all possible states from which the agent will make decisions.

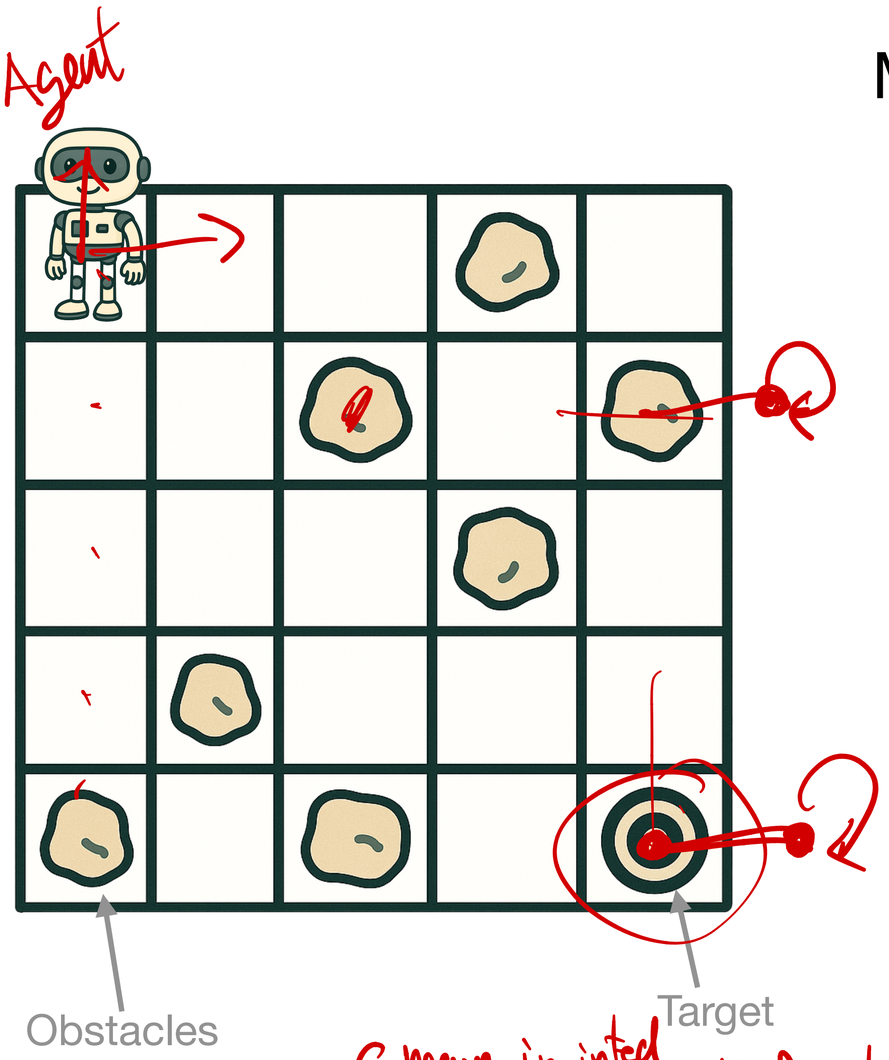
γ - Discount factor

$\gamma \in [0,1]$ — a modeling parameter that reduces the impact of future rewards on decisions made now.

T - Transition function

$s' \sim T(\cdot \mid s, a)$ a stochastic transition function defining the probability of the agent transitioning to next state $s' \in S$, given that it executed action $a \in A$ from state $s \in S$.

MDP Examples: Gridworld Navigation



MDP Components:

State space S — All possible locations of the agent in the environment.

$$S = \{(x, y) \mid x \in \{0,1,2,3,4\}, y \in \{0,1,2,3,4\}\}$$

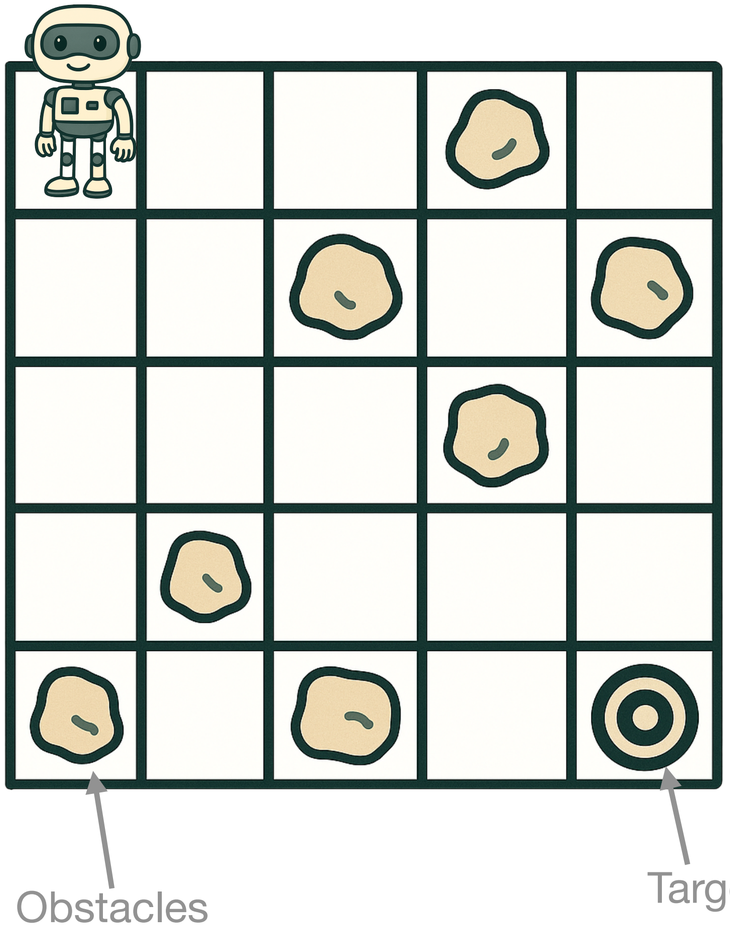
Action space A — Movement actions

$A = \{\text{left, right, up, down, diagonal, stay still}\}$

Transition function T — Move as planned/intended, so long as that action is valid, else, stay still. "Slip" with probability $p \in (0,1)$

$$T(s' \mid s, a) = \begin{cases} \text{move in inted direction} & \text{w.p. } 1-p \\ \text{move to random} & \text{w.p. } p \end{cases}$$

MDP Examples: Gridworld Navigation



MDP Components:

Reward function R Reward the agent for reaching the goal, and penalize for hitting an obstacle.

$$R(s,a) = \begin{cases} +1 & \text{if } s = (4,4) \\ -1 & \text{if } s = \{(3,0), (2,1), \dots\} \end{cases}$$

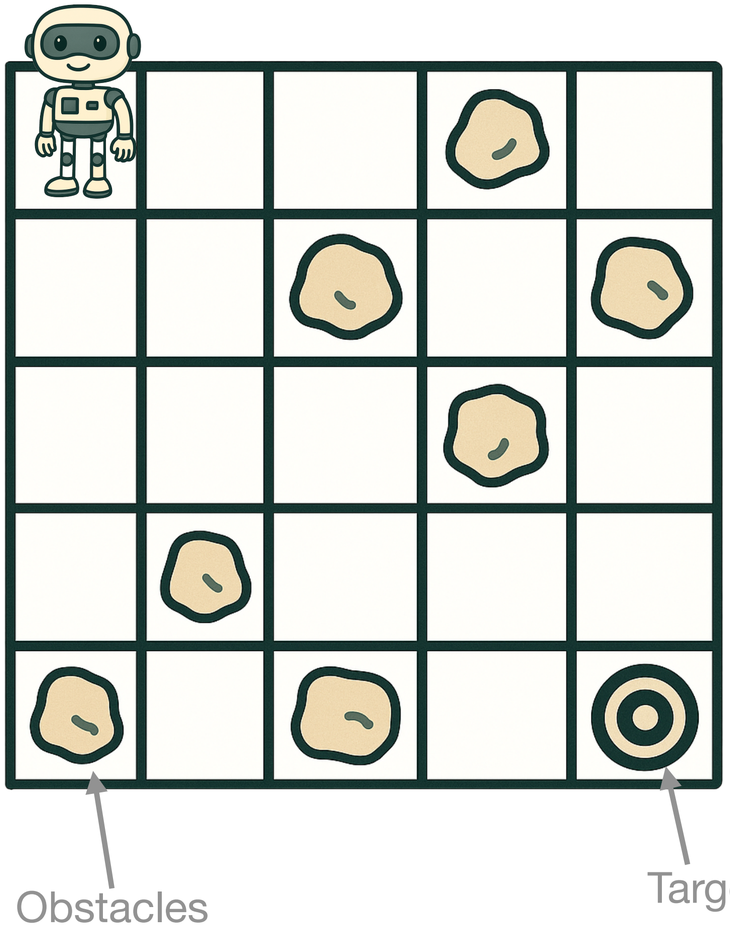
Initial distribution $\mu(s)$

We have options

$$\mu(s) = \begin{cases} 1 & \text{if } (0,0) = s_0 \\ 0 & \text{otherwise} \end{cases}$$

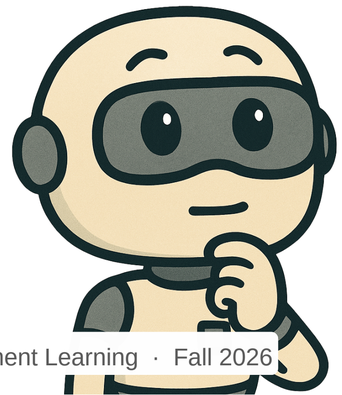
$\mu(s)$ = Uniform over states that can reach the goal.

MDP Examples: Gridworld Navigation



Why include stochastic transitions?
What could they model?

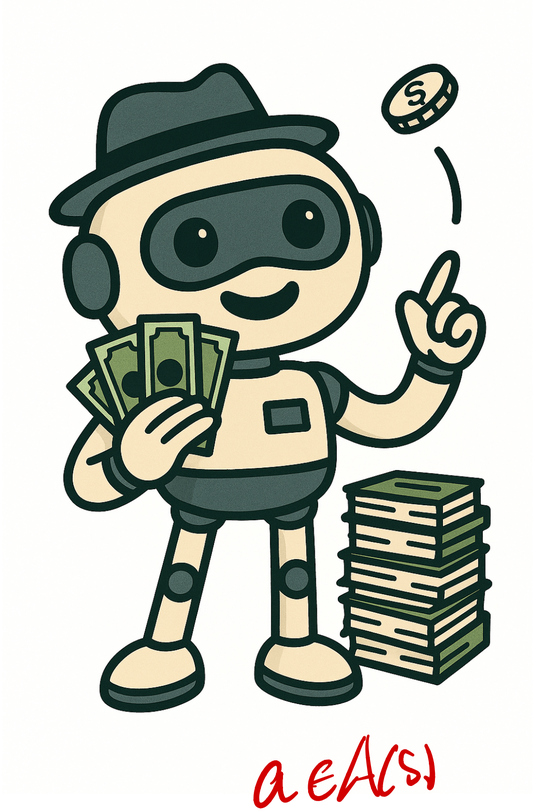
- Adding entropy to the problem that can help find the solution.
- When training it could help exploration
- Stochasticity can model uncertainty
  1. in environment transitions.
  2. Can also help make policies robust to observational uncertainty.



MDP Examples: Gambler's Ruin

$$T(s'|s,a) = \begin{cases} p & \text{if } s' = \min(\cdots) \\ 1-p & \text{if } s' = s-a \\ 0 & \text{otherwise} \end{cases}$$

- A gambler plays a game in which they make a bet of B dollars at every turn.
- They then have a probability p of winning $2B$ dollars, and a probability $1-p$ of losing the B dollars that they bet.
- They start with 50 dollars, and their goal is to cash out once they have have reached 100 dollars.



MDP Components:

State space S

The current amount of money the agent has

$$S = \{0, 1, 2, \cdots, 100\}$$

Action space A

They can bet an amount between 0 dollars and current state $s \in S$.

$$A(s) = \{0, 1, 2, \cdots, s\}$$

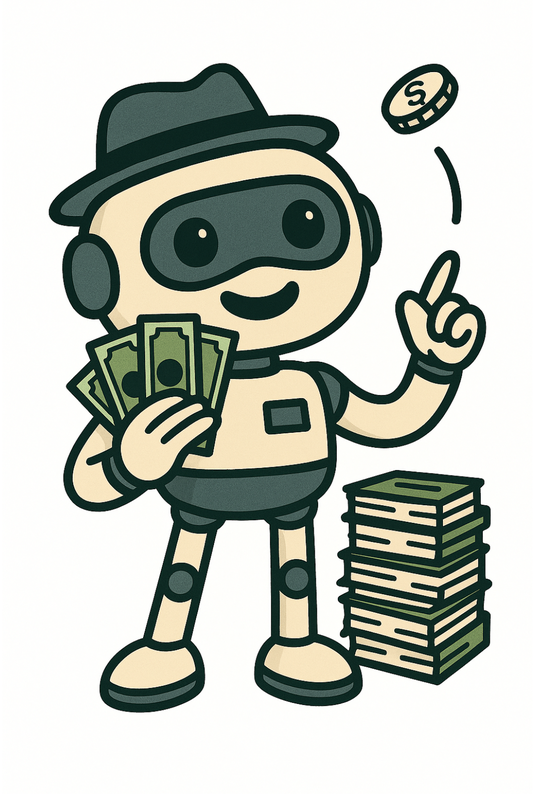
Transition function T

$$T(s'|s,a) = \begin{cases} s' = \min(100, s+a), & \text{w.p. } p \\ s' = s-a & \text{w.p. } 1-p \end{cases}$$

$a \in A(s)$

MDP Examples: Gambler's Ruin

- A gambler plays a game in which they make a bet of B dollars at every turn.
- They then have a probability p of winning $2B$ dollars, and a probability $1-p$ of losing the B dollars that they bet.
- They start with 50 dollars, and their goal is to cash out once they have have reached 100 dollars.



MDP Components:

Reward function R

Win 100 and cash out.

$$R(s,a,s') = \begin{cases} +1 & \text{if } s'=100 \\ 0 & \text{otherwise} \end{cases}$$

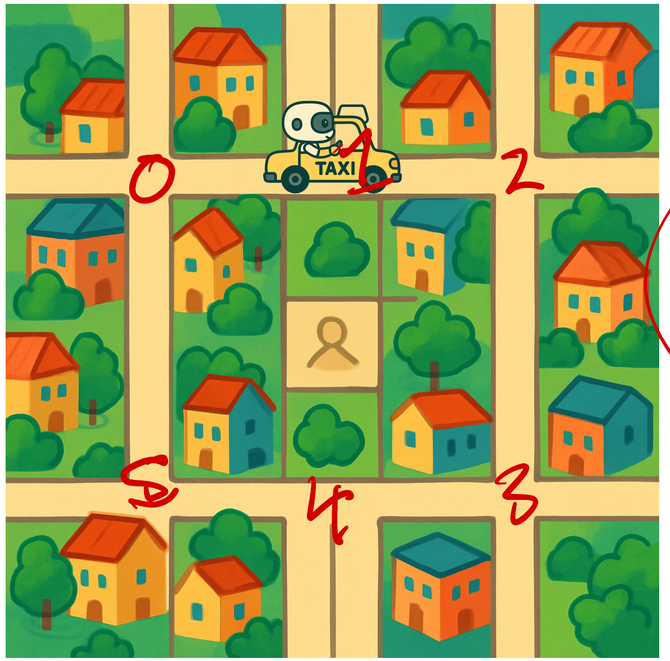
Initial distribution $\mu(s)$

$$\mu(s) = \begin{cases} 1 & \text{if } s_0 = 50 \\ 0 & \text{otherwise.} \end{cases}$$

Note: we <u>could</u> also define a distribution over various possible starting states.

MDP Examples: Autonomous Taxi Example

- An autonomous taxi needs to pickup and dropoff passengers in a city.
- Passengers randomly spawn at locations throughout the city, and each has a different dropoff location.
- The taxi can only carry at most 1 passenger at a time.
- The taxi gets $\$5$ each time they deliver a passenger to the dropoff point.



MDP Components:

passenger flag

State space S

$S = \{0,1,2,3,4,5\} \times \{\text{index of which passenger}\}$
$\times \{\text{passenger destinations}\}$
$\times \{\text{passenger locations}\}$

Action space A

$A = \{\text{move up, left, down, right, pickup, dropoff}\}$

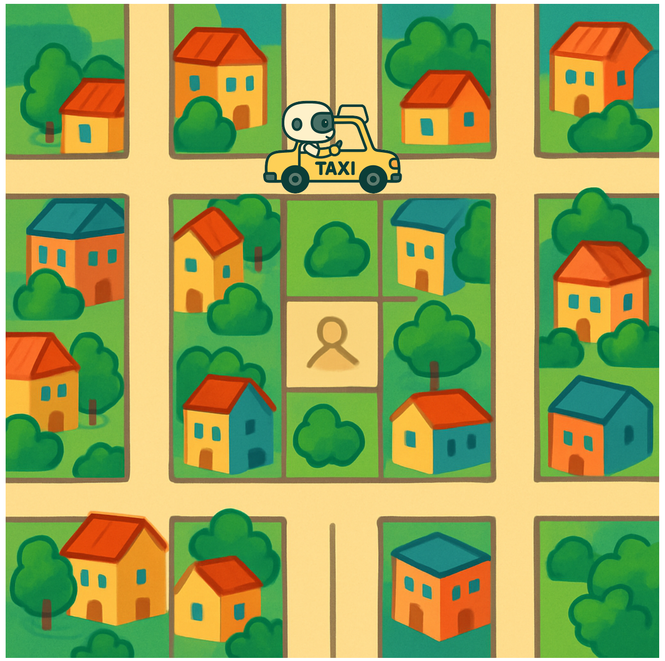
Transition function T

$T(s'|s,a)$

- Move from point to point
- Dropoff/pickup passengers according to action taken and the location of passengers.
- Spawn passengers randomly

MDP Examples: Autonomous Taxi Example

- An autonomous taxi needs to pickup and dropoff passengers in a city.
- Passengers randomly spawn at locations throughout the city, and each has a different dropoff location.
- The taxi can only carry at most 1 passenger at a time.
- The taxi gets $\$5$ each time they deliver a passenger to the dropoff point.



MDP Components:

Reward function R

$$R(s,a,s') = \begin{cases} +5 & \text{if dropoff in correct location} \\ 0, & \text{otherwise} \end{cases}$$

Initial distribution $\mu(s)$

- Random distribution for passenger locations and destinations
- Random distribution for taxi location.

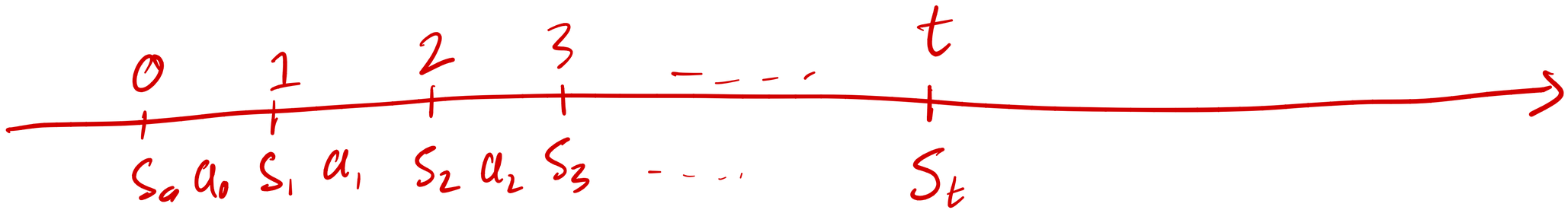
Decision Rules

A *decision rule* d_t is a function specifying an action to be executed at an individual decision epoch t.

Decision Rules

A *decision rule* d_t is a function specifying an action to be executed at an individual decision epoch t.

A decision rule is *history-dependent* if it relies on previous states and actions: $d_t(s_0, a_0, \ldots, s_t)$



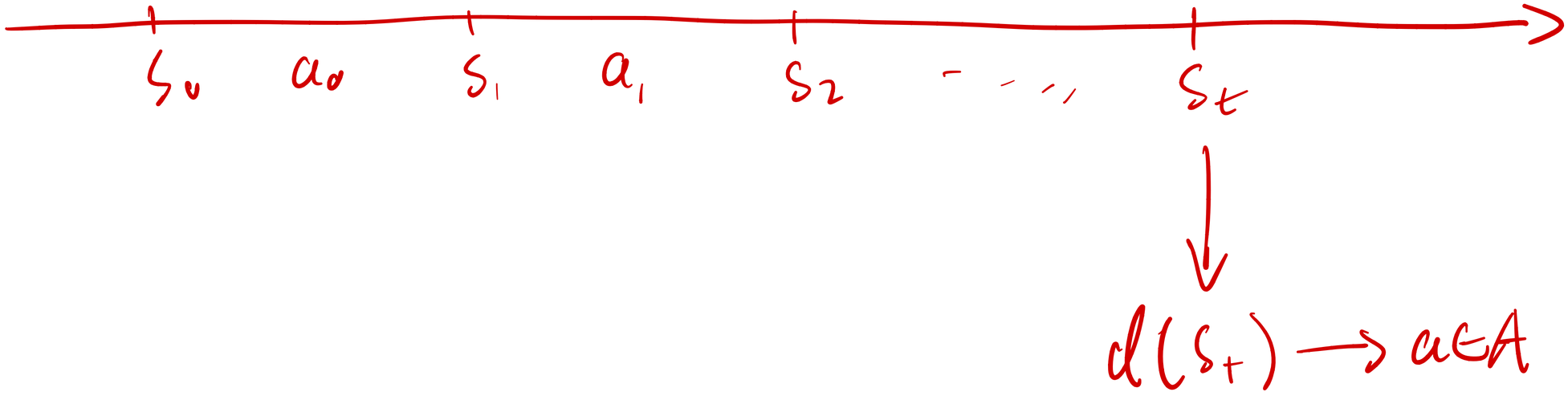
$$d_t(s_0, a_0, \ldots, s_t) \longrightarrow a \in A$$

Decision Rules

A *decision rule* d_t is a function specifying an action to be executed at an individual decision epoch t.

A decision rule is *history-dependent* if it relies on previous states and actions: $d_t(s_0, a_0, \ldots, s_t)$

A decision rule is *Markovian* if it only relies on the current state: $d_t(s_t)$



Decision Rules

A *decision rule* d_t is a function specifying an action to be executed at an individual decision epoch t.

A decision rule is *history-dependent* if it relies on previous states and actions: $d_t(s_0, a_0, \ldots, s_t)$

A decision rule is *Markovian* if it only relies on the current state: $d_t(s_t)$

A decision rule is *deterministic* if it outputs a single action $d_t(s_t) = a \in A$

Decision Rules

A *decision rule* d_t is a function specifying an action to be executed at an individual decision epoch t.

A decision rule is *history-dependent* if it relies on previous states and actions: $d_t(s_0, a_0, \ldots, s_t)$

A decision rule is *Markovian* if it only relies on the current state: $d_t(s_t)$

A decision rule is *deterministic* if it outputs a single action $d_t(s_t) = a \in A$

A decision rule is *random* if it outputs a distribution over actions $d_t(a \mid s) = \mathbb{P}(a_t = a \,|\, s_t = s)$

Decision Rules

A *decision rule* d_t is a function specifying an action to be executed at an individual decision epoch t.

A decision rule is *history-dependent* if it relies on previous states and actions: $d_t(s_0, a_0, \ldots, s_t)$

A decision rule is *Markovian* if it only relies on the current state: $d_t(s_t)$

A decision rule is *deterministic* if it outputs a single action $d_t(s_t) = a \in A$

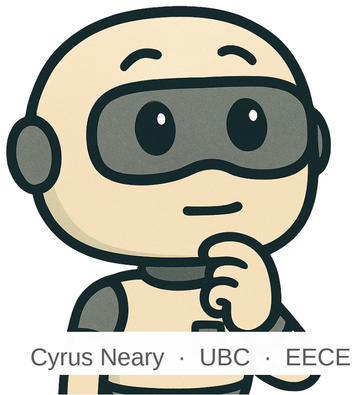
A decision rule is *random* if it outputs a distribution over actions $d_t(a \mid s) = \mathbb{P}(a_t = a \,|\, s_t = s)$

Decision rules can be:

	History-Dependent	**Markovian**
Deterministic	HD	MD
Random	HR	MR

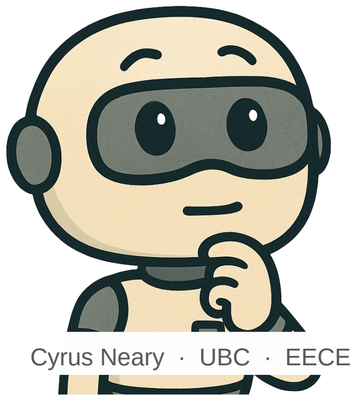
When would we expect a decision rule to need history in order to make good decisions?

- when the current observation is not sufficient to fully encode the state information necessary for decision making.
  - e.g. • Chat bot needs history/context to understand prompt.
    - digital filters.
- If the state is "properly" defined, we won't need history.



When might a random decision rule be helpful over a deterministic one?

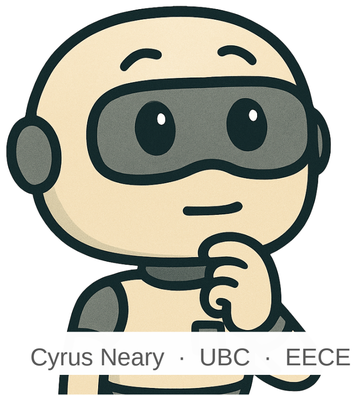
- In learning setting, we want to "explore". Random decision rule can help with this.
- In multiplayer/games, it can be necessary to have random decision rules to play optimally.
- Maybe it's easier to define/implement random decision rules.



More generally, why distinguish between different types of decision rules?

Each class of decision rules might have different algorithms for solving.

- we have to strike a balance between modeling fidelity (of the policy class) and computational burden of implementing the optimal policy.
  - How much info do we need to solve the problem optimally?
  - How much can we afford in terms of compute?



Policies and Policy Classes

A *policy* $\pi = d_0, d_1, \ldots$ is a sequence of decision rules that specifies which action the agent should take at **every** decision epoch.

policy π specifies

A policy belongs to a *policy class* $\Pi^{HD}, \Pi^{MD}, \Pi^{HR}, \Pi^{MR}$, if $\forall t,\ d_t \in D^{HD}, D^{MD}, D^{HR}, D^{MR}$, respectively.

e.g. $\pi \in \Pi^{MD}$ $d_0 \in D^{MD}, d_1 \in D^{MD}, \ldots, d_t \in D^{MD}$

A policy is *stationary* if the decision rule is constant in time, i.e. $d_t = d_{t'}$ for every t, t'.

i.e., only one decision rule d

In this course, we will often consider *stationary Markovian policies.*

Same decision rule at every timestep.
the decision rule is only a function of the current state.

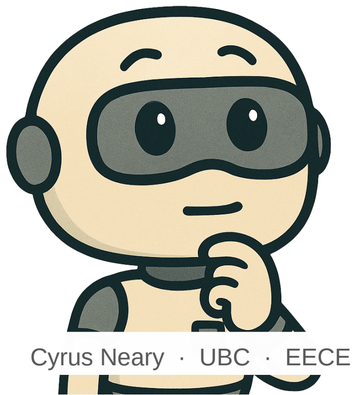
What's the most general class of policies?

We considered many classes of policies.

$$\Pi^{SD} \subset \Pi^{SR} \subset \Pi^{MR} \subset \Pi^{HR}$$

$$\Pi^{SD} \subset \Pi^{MD} \subset \Pi^{MR} \subset \Pi^{HR}$$

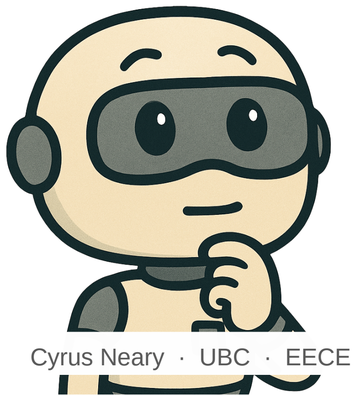
$$\Pi^{SD} \subset \Pi^{MD} \subset \Pi^{HD} \subset \Pi^{HR}$$



Why should we distinguish between stationary and non-stationary policies?

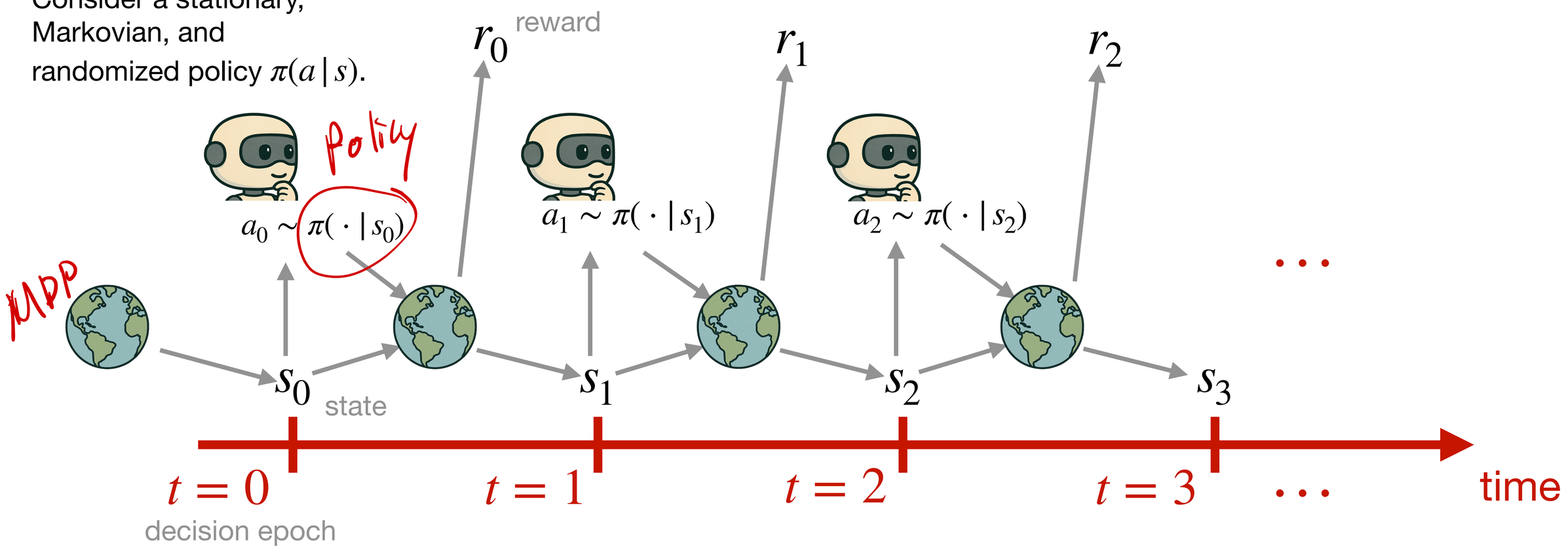
- Stationary policies are simpler, and necessary for infinite horizon problems.

- Non-stationary policies require lots of memory. Need to store separate decision rules for every timestep.



Executing a Policy in an MDP

Consider a stationary, Markovian, and randomized policy $\pi(a \mid s)$.



Executing a policy induces stochastic "trajectories" $\tau = s_0, a_0, r_0, s_1, a_1, r_1, \ldots$

Objective: choose $\pi(a \mid s)$ to maximize expectation of some function of trajectories.

$$\pi^* \in \text{argmax}_\pi \mathbb{E}[f(s_0, a_0, r_0, s_1, a_1, r_1, \ldots) \mid \pi]$$

Objective: Design “Performant” Policies w.r.t. Reward

A common optimization problem: Infinite-Horizon Total Discounted Reward

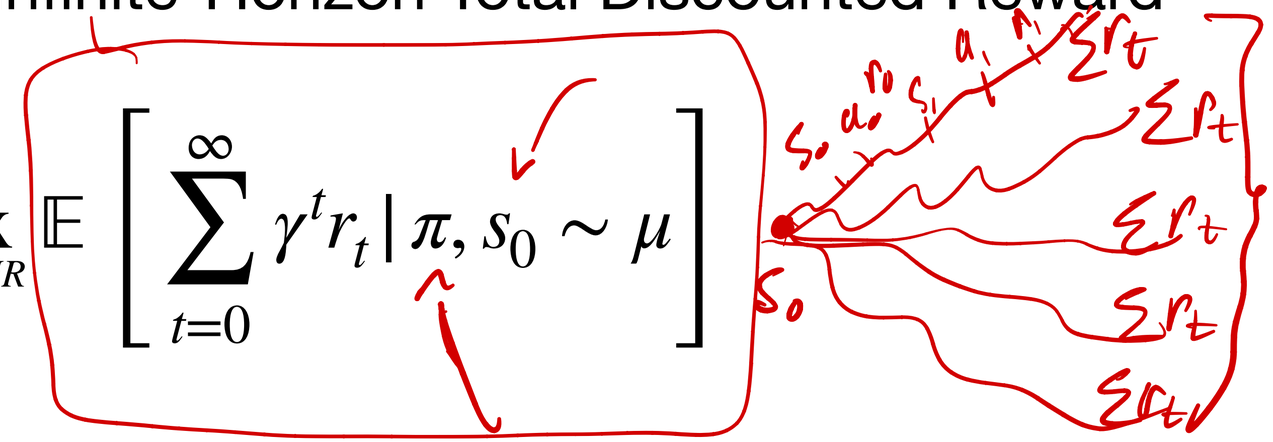
$$\pi^* \in \arg\max_{\pi \in \Pi^{HR}} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,|\, \pi, s_0 \sim \mu\right]$$

Intuition: Find a policy that maximizes the expected sum of discounted rewards.

Objective: Design “Performant” Policies w.r.t. Reward

A common optimization problem: Infinite-Horizon Total Discounted Reward

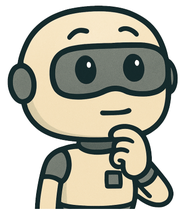
$$\pi^* \in \arg\max_{\pi \in \Pi^{HR}} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,|\, \pi, s_0 \sim \mu\right]$$



Intuition: Find a policy that maximizes the expected sum of discounted rewards.

What does this optimization problem actually mean?

- What distribution is the expectation taken over?
- Why is discounting necessary?
- When does such an optimal policy exist (particularly in a manageable policy class)?



Fixing a Policy in an MDP Induces Stochastic Processes

example with dice

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **finite time horizons**?

Ω Sample space, set of all possible outcomes $\omega \in \Omega$

e.g. for dice $\omega \in \{1,2,3,4,5,6\} = \Omega$

$\mathcal{F} \subseteq 2^\Omega$ "sigma algebra" of all possible "events".

↳ each event $A \in \mathcal{F}$, is a set of outcomes.

e.g. $A = \{2,4,6\}$ for "is the roll even"

$A = \{1\}$ for "did I role 1?"

$(\Omega, \mathcal{F}, \mathbb{P})$ — Probability space

$\mathbb{P} : \mathcal{F} \to [0,1]$ probability measure

e.g. $\mathbb{P}(A \text{ "roll even"}) = 1/2$

Random variable:

$X : \Omega \to \mathbb{R}$ is a measurable function w.r.t. probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^{\pi})$ for **finite time horizons**?

Sample space $\Omega = S \times A \times S \times A \times \ldots \times S = \{S \times A\}^{N} \times S$

$$\omega = (s_0, a_0, s_1, a_1, \ldots, s_N) \in \Omega$$

Each outcome ω is a potential *trajectory* of length N.

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **finite time horizons**?

Sample space $\Omega = S \times A \times S \times A \times \ldots \times S = \{S \times A\}^N \times S$

$$\omega = (s_0, a_0, s_1, a_1, \ldots, s_N) \in \Omega$$

Each outcome ω is a potential *trajectory* of length N.

σ-algebra $B(\Omega) = B(\{S \times A\}^{N-1} \times S)$

Events are sets of trajectories.

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **finite time horizons**?

Sample space $\Omega = S \times A \times S \times A \times \ldots \times S = \{S \times A\}^N \times S$

$$\omega = (s_0, a_0, s_1, a_1, \ldots, s_N) \in \Omega$$

Each outcome ω is a potential *trajectory* of length N.

σ-algebra $B(\Omega) = B(\{S \times A\}^{N-1} \times S)$

Events are sets of trajectories.

Probability measure

$$\begin{aligned}\mathbb{P}^\pi\left((s_0, a_0, s_1, a_1, \ldots, s_N)\right) &= \mu(s_0)\pi(a_0 \mid s_0)T(s_1 \mid s_0, a_0)\pi(a_1 \mid s_1)T(s_2 \mid s_1, a_1)\ldots \\ &= \mu(s_0)\prod_{t=0}^{N}\pi(a_t \mid s_t)T(s_{t+1} \mid s_t, a_t)\end{aligned}$$

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^{\pi})$ for **infinite time horions**?

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **infinite time horions**?

Sample space $\Omega = S \times A \times S \times A \times \dots$

$$\omega = (s_0, a_0, s_1, a_1, \dots) \in \Omega$$

Each outcome ω is a potential infinite *trajectory.*

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **infinite time horions**?

Sample space $\Omega = S \times A \times S \times A \times \dots$

$$\omega = (s_0, a_0, s_1, a_1, \dots) \in \Omega$$

Each outcome ω is a potential infinite *trajectory.*

σ-algebra $B(\Omega) = B(\{S \times A\}^\infty)$

Events are so-called cylinder sets.

Fixing a Policy in an MDP Induces Stochastic Processes

What's the probability space $(\Omega, B(\Omega), \mathbb{P}^\pi)$ for **infinite time horions**?

Sample space $\Omega = S \times A \times S \times A \times \dots$

$$\omega = (s_0, a_0, s_1, a_1, \dots) \in \Omega$$

Each outcome ω is a potential infinite *trajectory.*

σ-algebra $B(\Omega) = B(\{S \times A\}^\infty)$

Events are so-called cylinder sets.

Probability measure

Cylinder sets have probability mass. Infinite trajectories have measure zero.

$$\mathbb{P}^\pi\left((s_0, a_0, s_1, a_1, \dots, s_N)\right) = \mu(s_0)\pi(a_0 \mid s_0)T(s_1 \mid s_0, a_0)\pi(a_1 \mid s_1)T(s_2 \mid s_1, a_1)\dots$$

$$= \mu(s_0)\prod_{t=0}^{N} \pi(a_t \mid s_t)T(s_{t+1} \mid s_t, a_t)$$

Optimality Criterion Revisited

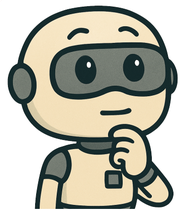
A common optimization problem: Infinite-Horizon Total Discounted Reward

$$\pi^* \in \arg\max_{\pi \in \Pi^{HR}} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,|\, \pi, s_0 \sim \mu\right]$$

Intuition: Find a policy that maximizes the expected sum of discounted rewards.

What does this optimization problem actually mean?

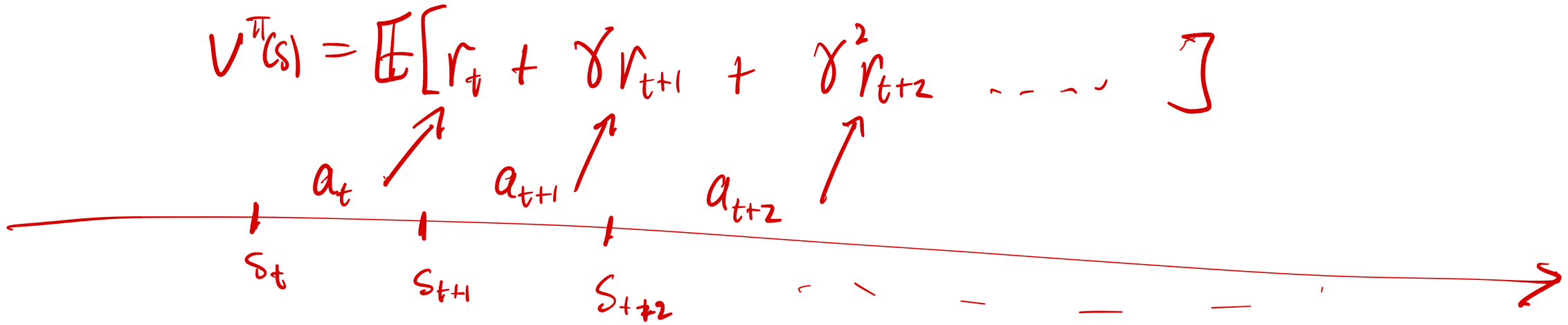
- What distribution is the expectation taken over?
- Why is discounting necessary?
- When does such an optimal policy exist (particularly in a manageable policy class)?



Defining the Value Function

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

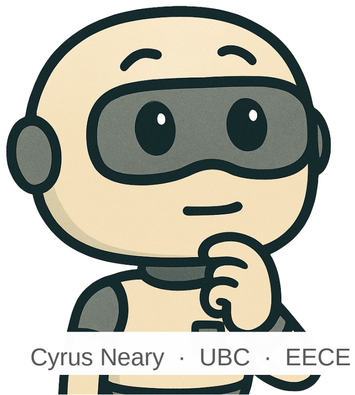
Intuition: How “valuable” is it for the agent to be in state s, if it follows policy π from that point onwards?



What Assumptions Did we Make When Defining the Value Function?

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

- We assumed that the infinite sum converges.
- We assumed $V^{\pi}(s)$ doesn't depend on the current time t.
  - ↳ This works for stationary policies and infinite time horizons and markovian states.



The Mathematical Importance of Discounting

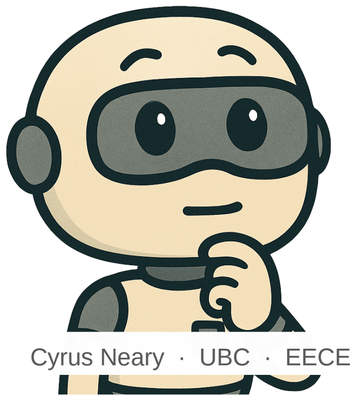
$$V^{\pi}(s) = \lim_{N \to \infty} \mathbb{E}\left[\sum_{t=0}^{N} \gamma^t r_t \mid \pi, s_0 = s \right]$$

When does the limit exist? Suppose $R(s,a)$ is bounded. $\exists M < \infty$ s.t. $|R(s,a)| \leq M$ $\forall s, a$

Consider $\lim_{N\to\infty} \mathbb{E}\left[\sum_{t=0}^{N} |\gamma^t r_t| \mid \pi, s_0\right] \leq \lim_{N\to\infty} \sum_{t=0}^{N} \gamma^t M$

$\gamma < 1$ $\lim_{N\to\infty} \sum_{t=0}^{N} \gamma^t M$ is a geometric series $\longrightarrow \frac{M}{1-\gamma}$

Series is absolutely convergent $\Rightarrow$ also convergent.



Why is this important? - Discounting is essential for the math of infinite horizon problems.

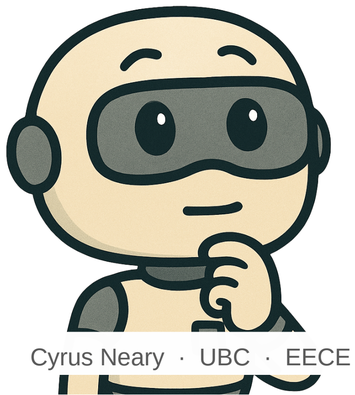
Otherwise, values could be infinite and things get tricky.

When is/isn't the Value Function Stationary in Time?

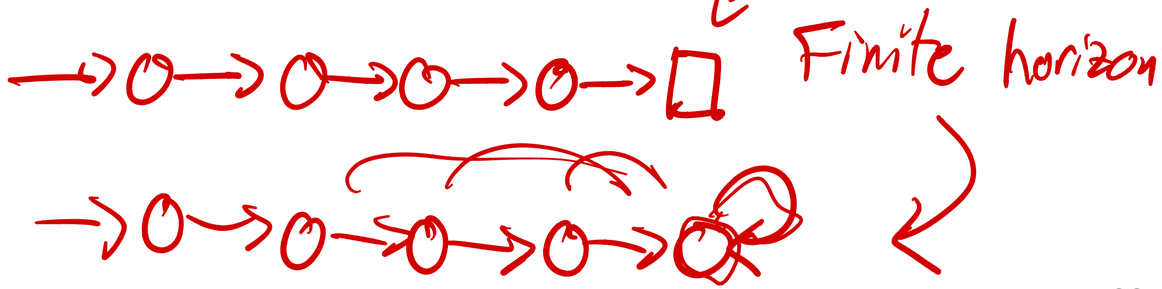
$$V^{\pi}(s) = \lim_{N \to \infty} \mathbb{E}\left[\sum_{t=0}^{N} \gamma^t r_t \mid \pi, s_0 = s \right]$$

- Stationary in time because MDP is stationary
  i.e. Dynamics, reward function, policy don't explicitly depend on time.

- It wouldn't necessarily be stationary if there was a finite time horizon.



Transforming finite horizon to infinite horizon.



Defining the Action-Value Function

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

Intuition: How “valuable” is it for the agent to be in state s, if it follows policy π from that point onwards?

Defining the Action-Value Function

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

Intuition: How “valuable” is it for the agent to be in state s, if it follows policy π from that point onwards?

$$Q^{\pi}(s, a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s, a_0 = a\right]$$

Intuition: How “valuable” is it for the agent to take action a from state s, if it follows policy π from that point onwards?

Defining the Action-Value Function

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s\right]$$

Intuition: How “valuable” is it for the agent to be in state s, if it follows policy π from that point onwards?

$$V^{\pi}(s) = \sum_{a \in A} \pi(a \mid s) Q^{\pi}(s, a) \quad \updownarrow \quad Q^{\pi}(s, a) = R(s, a) + \gamma \sum_{s' \in S} T(s' \mid s, a) V^{\pi}(s')$$

$$Q^{\pi}(s, a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \pi, s_0 = s, a_0 = a\right]$$

Intuition: How “valuable” is it for the agent to take action a from state s, if it follows policy π from that point onwards?

An Optimality Criterion: Infinite-Horizon Total Discounted Reward

A policy $\pi^*(\cdot \mid s)$ is considered optimal whenever

$$V^{\pi^*}(s) \geq V^{\pi}(s), \quad \forall s \in S, \forall \pi \in \Pi^{HR}$$

An Optimality Criterion: Infinite-Horizon Total Discounted Reward

A policy $\pi^*(\cdot \mid s)$ is considered optimal whenever

$$V^{\pi^*}(s) \geq V^{\pi}(s), \quad \forall s \in S, \forall \pi \in \Pi^{HR}$$

The optimal value of an MDP is defined as:

$$V^*(s) := \sup_{\pi \in \Pi^{HR}} V^{\pi}(s)$$

An Optimality Criterion: Infinite-Horizon Total Discounted Reward

A policy $\pi^*(\cdot \mid s)$ is considered optimal whenever

$$V^{\pi^*}(s) \geq V^{\pi}(s), \quad \forall s \in S, \forall \pi \in \Pi^{HR}$$

The optimal value of an MDP is defined as:

$$V^*(s) := \sup_{\pi \in \Pi^{HR}} V^{\pi}(s)$$

An optimal policy $\pi^* \in \Pi^{\{HR,MR,HD,MD\}}$ exists when:

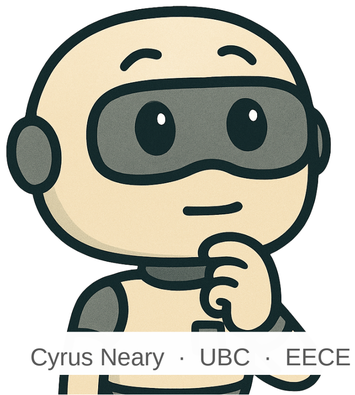
$$V^{\pi^*}(s) \geq V^*(s), \quad \forall s \in S$$

Optimality of Markovian Policies

What policy classes $\Pi^{HR}, \Pi^{HD}, \Pi^{MR}, \Pi^{MD}$ do we need to search to find an optimal policy?

- For any policy $\pi \in \Pi^{HR}$, for each s, there exists a policy $\pi' \in \Pi^{MR}$ for which $V^{\pi}(s) = V^{\pi'}(s)$
  Thm 5.5.3 in Puterman (for infinite horizon stationary MDPs)

- Also, there exists a policy $\pi \in \Pi^{MD}$ that is optimal (i.e. achieves at least as much value as any other policy).



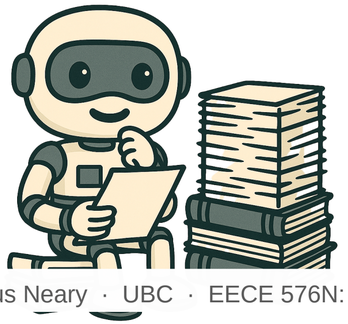
Other Optimality Criteria

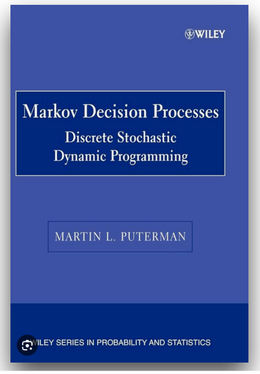
- So far, we've only presented one notion of optimality, assuming $0 \leq \gamma < 1$, and that our goal is to maximize the *sum* of future rewards.

- In this course, we will stick to discounted sums of reward over infinite horizons, due to the mathematical simplicity of this setting, and its dominance in the literature.

- Other notions of optimality in MDPs have been studied, and new ones can be defined (perhaps by you!).
  - E.g., Finite horizons, undiscounted rewards, average rewards, entropy-regularized value functions, risk-aware measures of value, etc.

- If you're interested in learning more, start by reading Puterman's book.





Chapters 4-10

Next Class: Solving for Optimal Policies